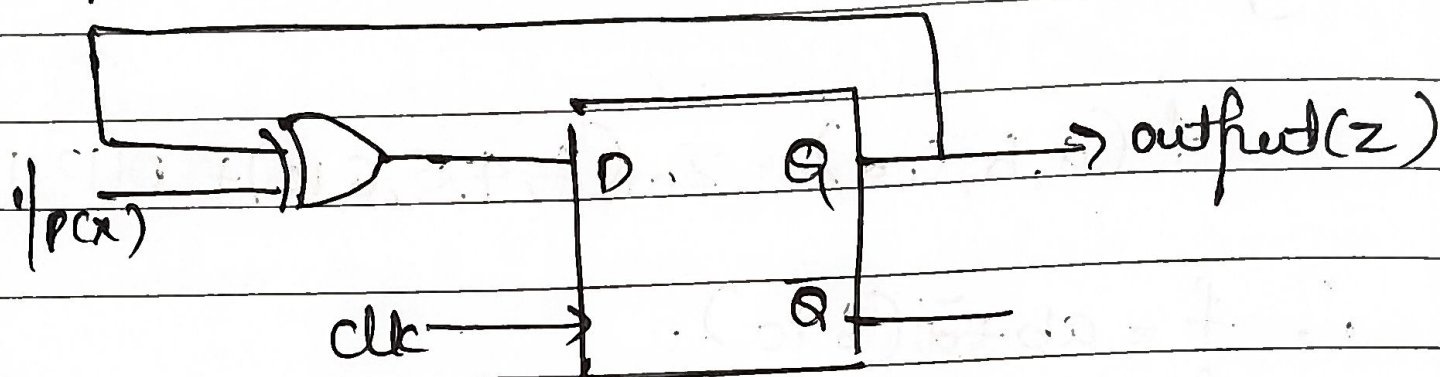


④ Sequential Assignment

1. Explain the behavior of the given circuit.



= The circuit contains D flipflop.

Q/p :- $D = x \oplus Q$

D flipflop transfers D to Q at every positive clock edge.

$$Q_{\text{next}} = D = x \oplus Q$$

$\therefore$

$$Z = Q$$

x

present Q

Next Q ($x \oplus Q$)

0

0

0

0

1

1

1

0

1

1

1

0

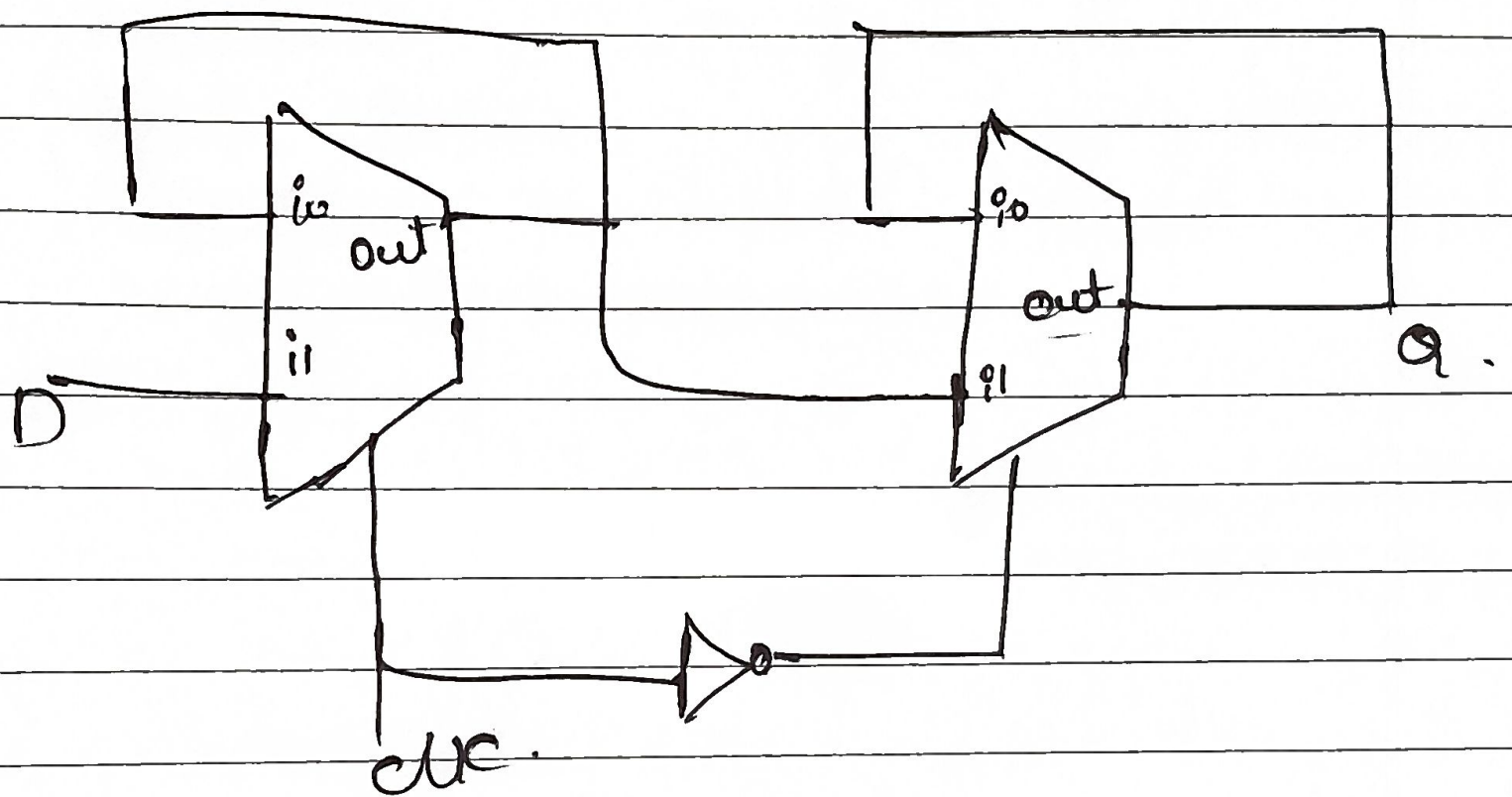
Behaviour:- when $x = 0$ $Q_{\text{next}} = Q$

when $x = 1$ $Q_{\text{next}} = \overline{Q}$

$\therefore$ The circuit behaves like a T flipflop

$$T = x$$

2. Design a p-edge triggered D flipflop using 2:1 Muxes only.

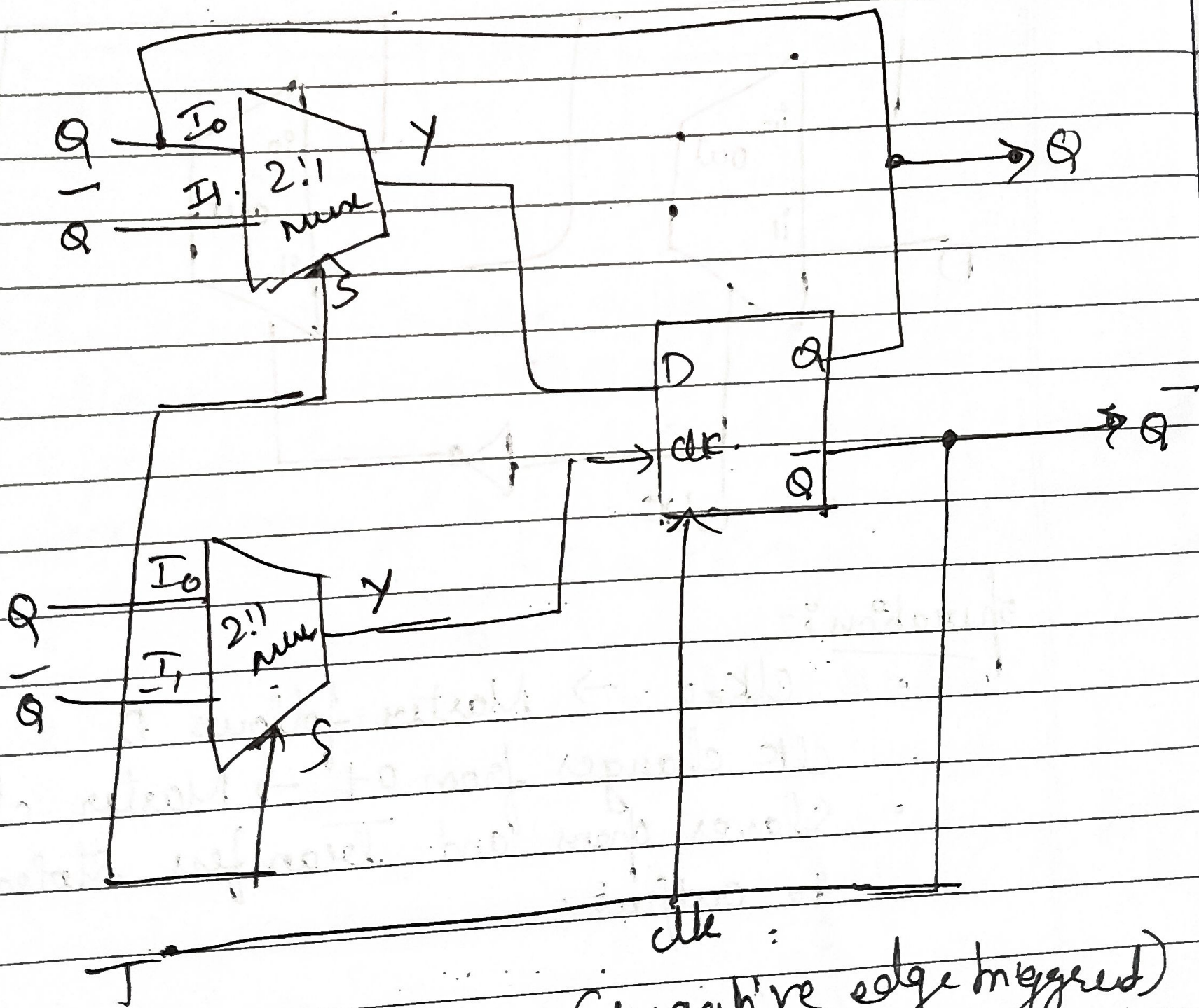


operation:-

- $clk = 0 \rightarrow$ Master follows D
- clk changes from 0 to 1 $\rightarrow$ Master closes
- Slaves opens and transfers stored data to output.

3. Design a Negative Edge triggered T flip-flop using only 2:1 Muxes.

$$D = T \oplus Q$$



(negative edge triggered)

4 Convert by adding external gates.

a) A D flipflop to a JK flipflop

b) A T flipflop to a D flipflop.

= a) Characteristics eqn of JK flipflop

$$Q_{n+1} = J\bar{Q}_n + \bar{K}Q_n.$$

for D FF, $Q_{n+1} = D.$

$$\therefore \underline{\underline{D = J\bar{Q} + \bar{K}Q}}$$

b) Characteristic eqn of T flipflop

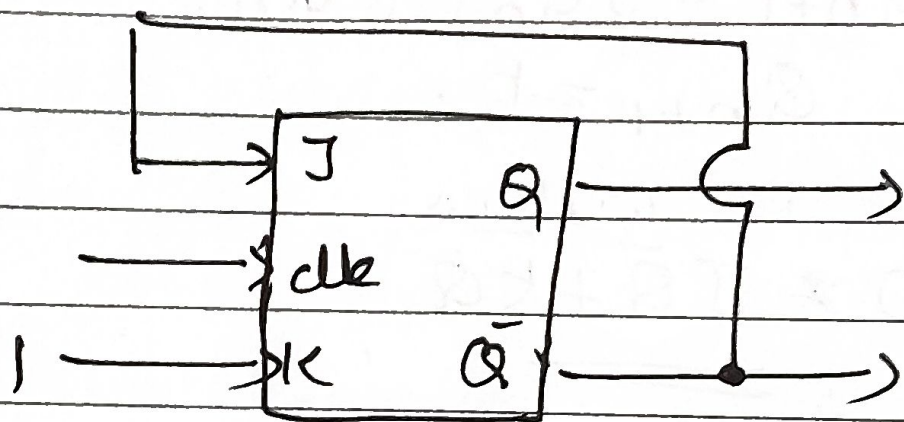
$$Q_{n+1} = T \oplus Q_n.$$

for D flipflop, $Q_{n+1} = D$

$$\therefore D = T \oplus Q$$

$$\therefore \underline{\underline{T = D \oplus Q}}$$

- 5) In a positive edge triggered JK flipflop, we have $J = Q'$ and $K = 1$ as shown in figure below. Assume the flipflop was initially cleared and then clocked for 6 clock pulses, find the sequence at the Q output for those 6 clock cycles.



= * Truth table of JK flipflop

J	K	Next State
0	0	no change
0	1	Reset (0)
1	0	Set (1)
1	1	Toggle

* Analyse each clock pulse

Initial State $Q_0 = 0$, $\bar{Q}_0 = 1$.

Clock pulse	Present Q	$J = \bar{Q}$	K	Next Q	
Initial	0	1	1	—	Given
1	0	1	1	1	toggle
2	1	0	1	0	Reset
3	0	1	1	1	toggle
4	1	0	1	0	Reset
5	0	1	1	1	toggle
6	1	0	1	0	Reset

∴ The o/p sequence including initial state
 $0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0 \rightarrow 1 \rightarrow 0$

After 6 clock pulse, the values of Q
 1, 0, 1, 0, 1, 0.

∴ Q sequence = 101010.